

ARIF UZ ZAMAN

AI/ML Architect & Technical Leader

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PROFESSIONAL SUMMARY

AI/ML Architect & Technical Leader driving enterprise data transformation and secure AI governance. Expert in engineering Medallion-structured data pipelines, orchestrating scalable multi-agent RAG ecosystems, and enforcing ISO 42001 compliance. Published author (M.Sc. BUET) with a proven track record of bridging complex machine learning execution with high-impact business strategy.

CORE SKILLS

AI Architecture & GenAI: PyTorch, TensorFlow, Llama 2/3, AWS Bedrock, LangChain, Qdrant, Neo4j, NLP.

MLOps & Engineering: GCP, Docker, Airflow, MLflow, CI/CD, GitLab, FastAPI, Python, W&B, Neptune.

Data & AI Governance: SQL, PgVector, ETL, ISO-42001, AI Security & Ethics, Power BI, Tableau, Superset.

Strategic Leadership: Product Strategy, Stakeholder Management, Arch. Design, Req. Engineering.

Programming Language: Python, Rust, Java and C++

PROFESSIONAL EXPERIENCE

BRAC IT – Business Intelligence Analyst, AI Product Delivery | Nov 2024 – Present

- **AI Governance Management System (BRAC IT & Tech):** Researched global AI regulations, developed PoCs
- **Legal Compass AI Chatbot (BRAC SELP):** Led requirement analysis, client meetings, mock development, and SRS creation; performed deep LLM system integration, AI governance design, data flow verification, and cloud/chatbot security analysis.
- **Payroll Data Migration:** Managed end-to-end database mapping, wrote migration & validation scripts, and implemented custom migration logic to guarantee source-to-destination data integrity.
- **Microfinance Biometrics Verification & Loan Processing (BRAC):** Directed client communications, SRS development, BA validation, and API integration testing; completed UAT with 50+ test cases and managed comprehensive risk mitigation.
- **BA Automation - Internal R&D (Agentic AI):** Built an agentic, RAG-based tool from scratch using **FastAPI, Neo4j, PostgreSQL, vector databases, and Ollama** to parse documents/images, ingest historical SRS/BRD files, and dynamically generate BA suggestions for internal use.
- **Data Pipeline & Data Governance (BRAC Tech DIYA):** Authored enterprise data governance reports and designed a Medallion data architecture (Bronze, Silver, Gold layers) alongside comprehensive data quality matrices.
- **Cloud Billing System (BRAC Tech):** Spearheaded initial system design, data validation, data flow definition, SRS development, stakeholder alignment, and risk analysis.
- **BRAC HR Chatbot (BRAC):** Conducted business system requirements analysis and code-level demo verification.
- **One BRAC AI Chatbot (BRAC International):** Performed business and system analysis, driving scalable Proof of Concept (PoC) development for global deployment.
- **DOTA Analytics & DOTA AI Platform (BRAC IT Global):** Developed user onboarding guides, built client relationships, and formulated business cases and tool verification documents for potential clients.
- **NOVA Reports:** Executed comprehensive business and technical evaluations to ensure full system accuracy and production readiness.
- **Hackathon - Code Sprint 2026:** Rapidly built and successfully deployed the "Odhikar" AI Legal Platform.

hSenid Business Solutions PLC – Implementation Engineer | Apr 2023 – Oct 2024

- Spearheaded 6 enterprise HRIS integrations, designing custom reporting architectures to support 1,000+ users across multinational business units.
- Standardized SQL-based SLA alerting systems and led technical training for enterprise teams across Sri Lanka, Maldives, and Indonesia.

BIZDATA INSIGHTS – Lead, Technology, Data & Business Development | Aug 2022 – Mar 2023

- Scaled engineering team performance through strategic mentorship, driving a 40% user acquisition surge and 25% retention lift for a core data terminal product.
- Strategized mission-critical data infrastructure for the Government of Bangladesh (in partnership with IQVIA India), enabling real-time pandemic analytics.

BIZDATA INSIGHTS – Specialist, Technology & Data | Aug 2021 – Aug 2022

- Engineered a scalable AWS NoSQL backend to launch Bangladesh's first FastAPI/Dash data terminal, driving a 37% increase in new customers
- Deployed OCR automation workflows, optimizing data pipelines and reducing manual data entry by 75%

Inovace Technologies – Project Engineer, IoT | Jul 2020 – Aug 2020 | IoT & R&D Intern | Jan 2020 – Jun 2020

TEACHING AND THOUGHT LEADERSHIP

Author | The Source of Reality (Amazon KDP) | 2026 | [\[LIVE LINK\]](#)

- Authored a comprehensive guide on the intersection of cosmology, theoretical physics and islamic religion

Lead Academy - AI & ML Course Instructor | Sep 2025 - Present

- Designing & instructing a comprehensive 32-class live and 16-recorded class advanced machine learning curriculum for industry professionals

Byte Science (YouTube Channel) - Creator and Educator | Jan 2026 - Present

- Research, Analyze and Develop videos on Science and Technology
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AI/ML PORTFOLIO & ARCHITECTURAL PROJECTS

- Multi-Agent Content Orchestrator: Architected a scalable, agentic framework (Gemini, Google Agent SDK) to fully automate complex content lifecycles and significantly reduce enterprise production overhead.
 - Auditable MLOps Ecosystem: Engineered a continuous fraud-analysis pipeline (DVC, MLflow, CI/CD) with real-time drift monitoring to guarantee model reliability and 100% auditability.
 - RAG Observability Platform: Deployed a containerized compliance system with real-time tracing to eliminate LLM opacity and rigorously enforce global safety standards.
 - Privacy-Centric Infrastructure: Scaled high-volume ETL pipelines (Airflow) processing 100K+ daily records and implemented secure semantic search using localized, containerized LLMs (Gemma3, Qdrant).
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OPEN SOURCE CONTRIBUTION

- **Superduper Framework** – Improved test reliability by fixing docstring validation logic in VectorIndex; passed 703/703 tests in CI suite.
 - **Oh-My-Openagent** - Patched a recurring execution failure in OpenCode to eliminate session-start errors, optimizing the prompt delivery pipeline for macOS users.
 - **Dagster** - Expanded Dagster's core execution engine to support DataVersionsByPartition in batched materializations, implementing robust type-propagation and strict invariant guards to resolve silent data-discard errors.
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EDUCATION

- M.Sc., Management of Technology – BUET, Bangladesh
 - B.Sc., Electronics and Communication Engineering – BRAC University, Bangladesh
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SELECTED CERTIFICATIONS (Full list available on LinkedIn)

- Certified AI Security and Governance from Securiti, Certification ID: 144E107B5-144E10624-1395F9FDD
 - Product Strategy Micro-Certification (PSC)[™] – Product School
 - Product Analytics Micro-Certification (PAC)[™] – Product School
 - Prompt Engineering from Beginner to Advanced, Certificate: [\[LIVE LINK\]](#)
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PUBLICATIONS

- Book: The Source of Reality: Unlocking the Hidden Code of the Universe | Amazon KDP | [\[LIVE LINK\]](#)
 - "Power Grid Classification through Electrical Network Frequency and its Multiple Frequency Components" – IEEE TENSYP 2020
 - "Crowdsourcing Medical Emergency System using Internet of Things in Bangladesh Perspective" – IEEE ICSIMA 2018
 - Book Chapter: *Toward an IoT-Based Solution for Emergency Medical System: An Approach to i-medical in Bangladesh* – Springer, 2021
 - Medium Publication: [\[PROFILE LINK\]](#)
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OTHER ACHIEVEMENTS

- Awarded ICT Scholarship in 2017
- Recognized as a Chess Master at BRAC IT

Mobility: Open to relocation to overseas in 10 - 16 weeks